

Unit 2 Pre-Test

Scientific Notation

Standards assessed: NC.8.EE.3, NC.8.EE.4

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.EE.3 Express very large and very small quantities in scientific notation and standard form.

- I can write numbers in scientific notation and in standard form, and recognize correct form.

Match each vocabulary term to its definition, then write each number as directed.

D ✓ scientific notation ✓	A. A number written the ordinary way, such as 45,000.
B ✓ coefficient ✓	B. The decimal factor in $a \times 10^n$ — at least 1 and less than 10.
A ✓ standard form ✓	C. Shows a small number between 0 and 1 (move the decimal left).
E ✓ base ✓	D. A number written as $a \times 10^n$, where $1 \leq a < 10$.
F ✓ positive exponent ✓	E. In a power of ten, the 10 that is multiplied repeatedly.
C ✓ negative exponent ✓	F. Shows a large number, 10 or greater (move the decimal right).

Parts of a number in scientific notation — reference

$$3.4 \times 10^5$$

coefficient	base	exponent
the decimal factor (1 to under 10)	always 10	how many places the decimal moves
A positive exponent makes a large number; a negative exponent makes a small one.		

1. Write in scientific notation

45,000

Answer:

answer: 4.5×10^4

2. Write in standard form

3×10^4

Answer:

answer: 30,000

3. Write in scientific notation

0.0062

Answer:

answer: 6.2×10^{-3}

4. Write in standard form

7.1×10^{-2}

Answer:

answer: 0.071

5. Write in scientific notation

580,000

Answer:

answer: 5.8×10^5

6. Fix the notation

Is 12.5×10^6 correct? Write the correct form.

Answer:

answer: 1.25×10^7

SCORE 4.0 Target Content

NC.8.EE.4 Perform operations with numbers in scientific notation and compare their sizes.

• I can multiply and divide numbers in scientific notation and tell how many times larger one is than another.

Multiply, divide, or compare. Write each answer in correct scientific notation.

7. Multiply

$(3 \times 10^4)(2 \times 10^3)$

Answer:

answer: 6×10^7

8. Divide

$(8 \times 10^9) \div (4 \times 10^5)$

Answer:

answer: 2×10^4

9. Multiply

10. Compare

$$(5 \times 10^6)(4 \times 10^3)$$

Answer:

answer: 2×10^{10}

11. Divide

$$(9 \times 10^{-2}) \div (3 \times 10^4)$$

Answer:

answer: 3×10^{-6}

How many times larger is 6×10^8 than 2×10^3 ?

Answer:

answer: 3×10^5

12. Multiply

$$(2.5 \times 10^5)(2 \times 10^{-8})$$

Answer:

answer: 5×10^{-3}

SCORE 5.0 Advanced Content

NC.8.EE.4 Add and subtract in scientific notation and solve multi-step real-world problems.

• *I can add and subtract in scientific notation and solve real-world problems, writing answers in scientific notation.*

Solve. Write each answer in correct scientific notation.

13. Add

$$(4 \times 10^5) + (3 \times 10^5)$$

Answer:

answer: 7×10^5

15. Subtract

$$(8.4 \times 10^{-3}) - (2.1 \times 10^{-3})$$

Answer:

answer: 6.3×10^{-3}

17. Apply

14. Add

$$(5 \times 10^6) + (3 \times 10^5)$$

Answer:

answer: 5.3×10^6

16. Apply

A proton's mass $\approx 1.67 \times 10^{-27}$ kg.

Find the mass of 2,000 protons.

Answer:

answer: 3.34×10^{-24} kg

18. Apply

Light travels 3×10^8 m/s. How far in 2×10^2 seconds?

Answer:

answer: 6×10^{10} m

A server holds 4.5×10^{12} bytes. Each video is 9×10^5 bytes. How many videos fit?

Answer:

answer: 5×10^6